

Example: $X_1 \dots X_n \stackrel{iid}{\sim} U(\theta, \theta+1)$

Goal: estimate θ

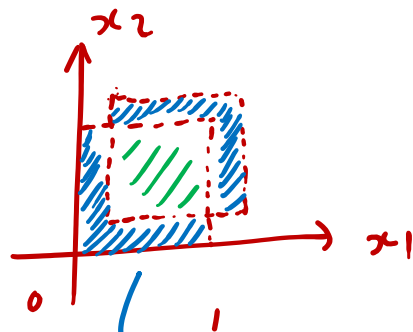
$$\inf_{\hat{\theta}} \sup_{\substack{\theta \in \Theta \\ \sim \\ \mathbb{R}}} \mathbb{E}[(\hat{\theta} - \theta)^2] \geq \underbrace{O\left(\frac{1}{n^2}\right)}_m$$

Sol:
$$m \geq \frac{\delta^2}{2} \left[1 - \|P_\theta^n - P_{2\delta}^n\|_{TV} \right]$$

$$p = P_\theta^n = U(\overset{\swarrow}{0}, \overset{\swarrow}{1}) \times \dots \times U(\overset{\swarrow}{0}, \overset{\swarrow}{1})$$

$$q = P_{2\delta}^n = U(2\delta, 1+2\delta) \times \dots \times U(2\delta, 1+2\delta)$$

$$\|p - q\|_{TV} = \frac{1}{2} \int |p - q| d\underline{x} \quad \checkmark$$



$$\|P_0^2 - P_{2\delta}^2\|_{TV} = 1 - (1 - 2\delta)^2$$



$$\|P_0^3 - P_{2\delta}^3\|_{TV} = 1 - (1 - 2\delta)^3$$

$$\delta [1 - 2\delta - n\delta] = 0 \quad \begin{cases} \delta = 0 \\ \delta = \frac{1}{n+2} \end{cases}$$

$$\|P_0^n - P_{2\delta}^n\|_{TV} = 1 - (1 - 2\delta)^n$$

$$\Rightarrow m \geq \frac{\delta^2}{2} [1 - 1 + (1 - 2\delta)^n] = \frac{\delta^2}{2} (1 - 2\delta)^n$$

$\delta(1 - 2\delta)^n - \delta^2 n (1 - 2\delta)^{n-1} = 0 \rightarrow$

$\delta(1 - 2\delta) - n\delta^2 = 0$

$\delta = \frac{1}{2}$

$$\rightarrow m \geq \underbrace{\frac{1}{2} \left(\frac{1}{n+2} \right)^2}_{O\left(\frac{1}{n^2}\right)} \underbrace{\left[1 - \frac{2}{n+2} \right]^n}$$

Decoupling for density metrics

$$D(P^n, q^n) = n D(P, q)$$

$$D_{KL}(P^n \parallel q^n) = n D_{KL}(P \parallel q)$$

$$m \geq \frac{\Phi(\delta)}{2} \left[1 - \parallel P_0^n - P_{2\delta}^n \parallel_{TV} \right] \geq \frac{\Phi(\delta)}{2} \left[1 - \sqrt{\underbrace{2D_{KL}(P^n \parallel q^n)}_{nD_{KL}(P \parallel q)}} \right]$$

Example: $\parallel P - q \parallel_{TV} \leq \sqrt{\frac{1}{2} D_{KL}(P \parallel q)} = \infty$

$$D_{KL}(P \parallel q) = \int p(x) \log \frac{p(x)}{q(x)} dx$$

$$\exists x \quad p(x) \neq 0 \\ q(x) = 0$$

Squared Hellinger Distance

$$H^2(p \parallel q) := \int (\sqrt{p} - \sqrt{q})^2 dx$$

① Decoupling

② Upper bound TV dis based on Hellinger

$$\int \sqrt{pq} dx = 1 - \frac{1}{2} H^2$$

$$\textcircled{1} \quad \frac{1}{2} H^2(p^n \parallel q^n) = 1 - (1 - \frac{1}{2} H^2(p \parallel q))^n$$

Proof: $H^2 = \int (p + q - 2\sqrt{pq}) dx = 2 - 2 \int \sqrt{pq} dx = 2 \left[1 - \int \sqrt{pq} dx \right]$

$$H^2(P^n \parallel q^n) = 2 \left[1 - \underbrace{\int \sqrt{p^n q^n} dx}_\downarrow \right]$$

$$\int \dots \int \sqrt{p(x_2)} \sqrt{q(x_2)} \left(\underbrace{\int \sqrt{p(x_1)} \sqrt{q(x_1)} dx_1}_{1 - \frac{1}{2} H^2(p \parallel q)} \right) dx_2 \dots dx_n$$

$$= \left[1 - \frac{1}{2} H^2(p \parallel q) \right]^n$$

$$\Rightarrow \frac{1}{2} H^2(P^n \parallel q^n) = 1 - \left(1 - \frac{1}{2} H^2(p \parallel q) \right)^n$$

$$(2) \quad \|p - q\|_{TV} \leq H(p \| q) \sqrt{1 - \frac{H^2(p \| q)}{4}}$$

proof :

$$\begin{aligned} & \frac{1}{2} \int |p - q| dx \leq \frac{1}{2} \sqrt{\int (\sqrt{p} - \sqrt{q})^2 dx} \cdot \sqrt{\int (\sqrt{p} + \sqrt{q})^2 dx} \\ & \quad \underbrace{(\sqrt{p} - \sqrt{q})(\sqrt{p} + \sqrt{q})}_{H(p \| q)} \\ & \quad \underbrace{\left[\int_{p > q} p - q + \int_{q > p} q - p \right] / 2}_{\int |a(x)| \cdot |b(x)| dx} \end{aligned}$$

$$\int |a(x)| \cdot |b(x)| dx \leq \sqrt{\int a^2(x) dx} \sqrt{\int b^2(x) dx}$$

$$\int (p + q + 2\sqrt{pq}) dx = 2 + 2 \int \sqrt{pq} dx = 2 \left[1 + \underbrace{\int \sqrt{pq} dx}_{1 - \frac{1}{2} H^2} \right]$$

$$\|p - q\|_{TV} \leq \frac{1}{2} H(p \| q) \sqrt{2(2 - \frac{1}{2} H^2)} = H(p \| q) \sqrt{1 - \frac{1}{4} H^2} \leq H(p \| q)$$

$$\|P^n - q^n\|_{TV} \leq H(P^n \| q^n) = 2 \sqrt{1 - (1 - \frac{1}{2} H^2(P \| q))^n}$$

$$\leq 2 \sqrt{\frac{1}{2} n \underbrace{H^2(P \| q)}_{\uparrow 4\delta}} = \cancel{2}$$

$$H^2(U(0, +1) \| U(2\delta, 1+2\delta)) = \int (\sqrt{p} - \sqrt{q})^2 dx$$

$$= \int_0^{2\delta} dx + \int_1^{1+2\delta} dx = 4\delta$$

$$2\delta = \frac{1}{4n}$$

Examp 15.5

$$\|P^n - q^n\|_{TV} \leq 2 \sqrt{\frac{1}{2} n}$$

$$n \geq \frac{S^2}{2} [1 - \|P_0^n - P_{2\delta}^n\|_{TV}] \geq \underbrace{6 \left(\frac{1}{n}\right)^2}$$

$$\delta = O\left(\frac{1}{n}\right)$$